

KOLDING VAMDRUP AP, DenmarkWMO: **061080**Lat: **55.438N**Lon: **9.334E**Elev: **44**StdP: **100.80**Time Zone: **1.00 (EUC)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-5.8	-11.0	1.5	-6.0	-8.5	1.8	-4.5	20.5	5.5	17.0	6.4	2.4	320	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	26.5	18.5	24.8	17.8	22.9	16.8	19.5	24.4	18.6	22.9	17.7	21.5	4.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	12.9	21.5	17.0	12.2	20.3	16.1	11.5	19.3	55.7	24.5	52.6	23.0	50.0	21.6	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	12.3	10.4	-12.8	28.9	4.6	1.8	-16.1	30.3	-18.8	31.3	-21.3	32.4	-24.7	33.7		
(5)			-13.4	21.4	4.4	1.0	-16.5	22.1	-19.1	22.7	-21.5	23.2	-24.7	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	1.2	1.5	3.7	7.9	11.5	14.3	16.5	16.6	13.7	9.4	5.3	3.0
	DBStd	6.39	3.91	3.46	3.25	3.16	3.14	2.82	2.83	2.59	2.57	3.08	3.29	4.16
	HDD10.0	1213	272	237	194	78	19	2	0	0	2	47	144	218
	HDD18.3	3527	531	470	452	313	214	124	73	66	140	276	392	476
	CDD10.0	766	0	0	0	15	65	132	202	206	114	30	2	0
	CDD18.3	38	0	0	0	0	1	4	17	14	2	0	0	0
	CDH23.3	306	0	0	0	1	8	46	151	93	7	0	0	0
	CDH26.7	39	0	0	0	0	0	6	23	11	0	0	0	0
Wind	WSAvg	4.9	5.5	5.9	5.6	5.1	5.0	4.8	4.0	4.1	4.2	4.6	4.9	5.1
Precipitation	PrecAvg	860	80	62	53	41	52	69	73	95	85	92	76	84
	PrecMax	1081	162	141	110	108	92	141	163	196	175	214	150	177
	PrecMin	594	7	17	9	5	10	2	11	34	18	39	17	22
	PrecStd	126	37	28	28	22	23	35	38	41	42	34	34	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	10.2	14.1	21.4	24.2	27.6	29.2	28.1	24.3	17.9	13.9	11.2
		MCWB	8.6	7.4	8.5	13.7	16.5	18.5	19.2	18.9	17.5	15.0	12.6	10.1
	2%	DB	8.2	8.4	11.2	17.9	22.0	24.6	26.9	25.9	21.9	16.0	12.2	10.1
		MCWB	7.4	7.1	7.6	11.3	14.6	17.8	18.6	18.4	16.3	13.6	11.1	9.4
	5%	DB	7.3	7.3	9.2	15.7	19.9	22.3	25.0	24.0	19.9	14.9	11.1	9.2
		MCWB	6.4	6.3	6.7	10.5	13.9	16.3	18.0	17.3	15.3	13.0	9.9	8.4
	10%	DB	6.4	6.4	8.0	13.8	17.9	20.2	22.9	22.0	18.2	13.9	10.0	8.2
		MCWB	5.5	5.4	6.4	9.6	12.9	15.3	17.0	16.8	14.6	12.2	8.9	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	8.2	9.7	14.8	18.0	20.4	21.0	20.4	18.5	15.5	12.9	10.3
		MCDB	9.3	9.1	12.0	20.4	22.6	25.0	26.0	25.4	22.8	17.2	13.7	10.9
	2%	WB	7.5	7.3	8.2	12.3	16.1	18.6	19.7	19.1	17.1	14.2	11.5	9.5
		MCDB	8.0	8.1	10.0	16.2	19.9	23.2	25.0	23.3	20.4	15.5	12.2	9.9
	5%	WB	6.5	6.5	7.3	10.9	14.5	17.1	18.7	18.3	16.1	13.2	10.2	8.6
		MCDB	7.1	7.3	8.8	14.6	18.6	21.2	23.5	22.2	18.8	14.4	11.0	9.1
	10%	WB	5.6	5.5	6.4	9.8	13.2	15.9	17.7	17.6	15.4	12.5	9.1	7.5
		MCDB	6.3	6.2	7.8	12.9	17.2	19.2	21.5	21.0	17.7	13.7	9.8	8.1
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.5	6.3	8.4	9.1	9.0	9.2	9.0	7.9	6.0	4.9	4.3
		MCDBR	4.1	5.0	8.4	11.9	12.9	12.6	12.7	11.9	10.3	6.9	4.9	3.8
		MCWBR	4.2	4.1	5.5	6.7	7.1	7.0	5.9	5.5	5.7	5.1	4.5	3.8
		MCDDBR	4.0	4.7	7.1	10.6	11.2	11.0	11.2	10.0	8.4	6.0	4.7	3.8
	5% WB	MCWBR	4.2	4.2	5.2	6.4	6.8	6.9	5.9	5.4	5.1	5.0	4.6	3.9
Clear-Sky Solar Irradiance	taub	0.320	0.339	0.364	0.397	0.389	0.384	0.400	0.400	0.378	0.356	0.334	0.307	
	taud	2.248	2.285	2.273	2.231	2.309	2.347	2.321	2.332	2.379	2.383	2.308	2.241	
	Ebn at Noon	584	724	796	821	853	861	837	811	778	698	566	507	
	Edh at Noon	58	79	101	121	119	116	117	109	91	72	54	46	
All-Sky Solar Radiation	RadAvg	0.49	1.11	2.43	4.05	5.28	5.57	5.21	4.17	2.89	1.50	0.61	0.36	
	RadStd	0.08	0.15	0.22	0.47	0.52	0.43	0.55	0.30	0.30	0.18	0.09	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-157	N/A	N/A

Nomenclature: See separate page